CHAPTER FIVE
INFORMAL FALLACIES

5.1. Fallacies in General

- Etymologically, the word fallacy derives from the Latin word "*fallere*" meaning "to deceive, to trip, to trick and to lead into error."
- The word may also derive from the Greek "*phelos*" meaning "deceitful."
- Fallacies are defect in an argument that consists in something other than merely false premises.
- Fallacies involve either a mistake in reasoning or the creation of some illusion that makes a bad argument appear good (or both).They involve motive to deceive or innocent mistake.

- They have deceptive appearance.
- They are deceptively persuasive. They are hard to spot and detect.
- Both deductive and inductive arguments may contain fallacies; if they do they are either unsound or uncogent, depending on the kind of the argument.
- Conversely, if an argument is unsound or uncogent, it has one or more false premises or it contains a fallacy (or both).

- Fallacies are usually divided into two groups: **formal** and **informal**.

- A **formal** fallacy is one that may be identified through mere inspection of the *form* or *structure* of an argument.

- Fallacies of this kind are usually found only in deductive arguments that have clearly recognizable forms.

Examples:

1. *If **x** then **y**.*
   *__**y**.__*
   *Therefore, **x**.*

2. *If **x** then **y**.*
   *__Not **x**.__*
   *Therefore, Not **y**.*

3. *All **x** are **z**.*
   *__All **y** are **z**.__*
   *So, all **x** are **y**.*

4. *All **x** are **y**.*
   *__No **z** are **x**.__*
   *So, no **z** are **y**.*

5. *Either **x** or **y**.*
   *__**x**.__*
   *Therefore, not **y***

6. *Either **x** or **y**.*
   *__**x**.__*
   *Therefore, **y**.*

Informal fallacies are those that can be detected only through analysis of the *content* of the argument.

Examples for Informal fallacy:

1. Everything that runs has feet. Nile River runs very swiftly. Therefore, Nile River has feet.

2. This dam is made of atoms. Atoms are invisible. Therefore, the dam is invisible.

Informal fallacies are classified into **five** major divisions:

I. Fallacies of Relevance
II. Fallacies of Weak Induction
III. Fallacies of Presumption
IV. Fallacies of Ambiguity
V. Fallacies of Grammatical analogy

5.2. Fallacies of Relevance

- There are **eight** fallacies under this category.
- They lack logical relevance.
- So, when an argument is psychologically or emotionally persuasive but logically incorrect, it contains fallacies of relevance.

1. Appeal to Force
(Appeal to the Stick)

- In this fallacy the arguer threatens the reader or listener that some psychological or physical harm will come to them unless they accept his or her conclusion. S/he threatens the listener with direct physical assault.

Examples:

1. *Dear professor, when considering my grade in your class keep in mind that my father is the dean, your boss.*
2. *I deserve a raise in salary for the coming year. After all, you know how friendly I am with your wife, and I'm sure you wouldn't want her to find out what's been going on between you and that sexpot client of yours.*

2. Appeal to Pity

- This fallacy also focuses on rousing the emotions of the reader or listener with the aim of gaining acceptance to one's conclusion.
- It attempts to evoke pity from the reader or listener.

Example:

Your Honor, I admit that I declared thirteen children as dependents on my tax return, even though I have only two. But if you find me guilty of tax evasion, my reputation will be ruined. I'll probably lose my job, my poor wife will not be able to have the operation that she desperately needs, and my kids will starve. Surely I am not guilty.

3. Appeal to People

- This fallacy tries to exploit our common psychological human desire i.e. the desire to be loved, esteemed, valued, recognized, admired and respected by a group in which the reader or listener is a member of or aspires to become a member of.
- It uses these desires to get the reader or listener to accept a conclusion.
- The assumption is that the reader wouldn't want to be left out of the group and risk all those desires by rejecting the arguer.

- The arguer tries to use and abuse that feeling of insecurity or need for security.
- Appeal to the people has two approaches: ***Direct and Indirect***.
- The *direct approach* occurs when an arguer, addressing a large group of people, excites the emotions and enthusiasm of the crowd to win acceptance for his or her conclusion.
- The objective is to arouse a kind of mob mentality.

- In the *indirect approach* the arguer aims his or her appeal not at the crowd as a whole but at one or more individuals separately, focusing on some aspect of their relationship to the crowd.
- The indirect approach includes such specific forms as:
  - the **bandwagon** argument, -majority
  - the appeal to **vanity -** popular
  - the appeal to **snobbery -** best

- Both the direct and indirect approaches of the *appeal to the people* fallacy have the same basic structure:

  "*You want to be accepted/included in the group/loved/esteemed. . . . Therefore, you should accept XYZ as true.*"

Examples:

1. *You should get a driver's license because everyone does.*
2. *Of course you want to buy these jeans it is the number one choice of people like Messy.*
3. *This traditional dress is not for everyone, it is only for proud genuine Ethiopian.*

4. Argument Against the Person

- It is committed when we ignore the substance of our opponent's argument and attack the arguer himself.
- It occurs in three forms:
  - ***Ad hominem* abusive –** to verbally abuse the opponent
  - ***Ad hominem* circumstantial -** the respondent attempts to discredit the opponent's argument by alluding to certain circumstances that affect the opponent. By doing so the respondent hopes to show that the opponent is predisposed to argue the way he or she does and should therefore not be taken seriously.
  - It takes the form *"Of course Mr. X argues this way; just look at the circumstances that affect him."*
  - **"You too" or *tu quoque*** attempts to make the opponent appear to be hypocritical or arguing in bad faith.

Examples:

1. *Einstein couldn't be right about Relativity for just look at the way he combs his hair.*
2. *We can't accept what Mr. x says about pork being unclean. After all he is a Jew.*
3. *Mr. 'x' has argued about the need to preserve family values. But who is he to talk? He has been married three times. He divorced his first wife while she was hospitalized for cancer, and he engaged in an extramarital affair while he was married to his second wife. Clearly, his arguments are trash.*

5. Accident

- The fallacy of **accident** is committed when a general rule is applied to a specific case it was not intended to cover. Typically, the general rule is cited (either directly or implicitly) in the premises and then wrongly applied to the specific case mentioned in the conclusion.

Examples:

1. *You shall not kill; hence, you shouldn't try to control termites in your home or fight for your country.*
2. *Everyone has the right to their own property. Thus, Jones, though he's been declared insane, shouldn't have had his gun taken from him.*

6. Straw Man

- The **straw man** fallacy is committed when an arguer distorts an opponent's argument for the purpose of more easily attacking it, demolishes the distorted argument, and then concludes that the opponent's real argument has been demolished.

Examples:

1. *Mr. 'x' has argued against prayer in the public schools. Obviously Mr. 'x' advocates atheism. But atheism is what they used to have in Russia. Atheism leads to the suppression of all religions and the replacement of God by an omnipotent state. Is that what we want for this country? I hardly think so. Clearly his argument is nonsense.*

2. ***X***: *"This novel is too hard for grade 12."*
   Y: *"I suppose you think that they should be reading comics."*

7. Missing the Point *(Ignoratio elenchi)*

- It occurs when the premises of an argument support one particular conclusion, but then a different conclusion, often vaguely related to the correct conclusion, is drawn.
- What do the premises logically imply?
- *Ignoratio elenchi* means "ignorance of the proof." The arguer is ignorant of the logical implications of his or her own premises and, as a result, draws a conclusion that misses the point entirely.

Examples:

1. *Crimes of theft and robbery have been increasing at an alarming rate lately. The conclusion is obvious: We must reinstate the death penalty immediately.*
2. *The quality of education is getting worse time after time. We should close all the schools.*

8. Red Herring

- It is committed when the arguer diverts the attention of the reader or listener by changing the subject to a different but sometimes subtly related one. He or she then finishes by either drawing a conclusion about this different issue or by merely presuming that some conclusion has been established.

Examples:

1. *There is a good deal of talk these days about the need to eliminate pesticides from our fruits and vegetables. But many of these foods are essential to our health. Carrots are an excellent source of vitamin A, broccoli is rich in iron, and oranges and grapefruit have lots of vitamin C.*
2. *I can't believe you thought that latest Disney movie was ok for children to watch. Disney pays 12-year old girls 31 cents an hour to sow their products together.*

5.3. Fallacies of Weak Induction

- they provide inadequate evidence to support their conclusion.
- the connection between premises and conclusion is not strong enough to support the conclusion.
- the premises provide at least a shred of evidence in support of the conclusion, but the evidence is not nearly good enough to cause a reasonable person to believe the conclusion.

9. APPEAL TO UNQUALIFIED AUTHORITY

- It occurs when the cited authority or witness lacks credibility.
- There are several reasons why an authority or witness might lack credibility. The person might lack the requisite expertise, might be biased or prejudiced, might have a motive to lie or disseminate "misinformation," or might lack the requisite ability to perceive or recall.

- Two key points to keep in mind:
  - One can be an expert in more than one area.
  - No one is authoritative when it comes to politics, morality , and religion.

Examples:

1. *Prof. Admasu, a philosopher, said that one couldn't catch Ebola by swimming together. Thus, you can't get infected with Ebola virus by swimming together.*
2. *The leader of KKK said African Americans were not good Americans. So, they are not good Americans.*
3. *The CEO of Tobacco Company argued that tobacco is not an addictive substance and that smoking cigarettes does not produce any addiction. So, smoking does not lead to any addiction.*
4. *Artist Yerdaw Tena said he saw the defendant stabbing the deceased several times. So, he is guilty.*

10. APPEAL TO IGNORANCE

- When the premises of an argument state that nothing has been proved one way or the other about something, and the conclusion then makes a definite assertion about that thing, the argument commits an **appeal to ignorance**.

Examples:

1. *Of course telepathy and other psychic phenomena do not exist. Nobody has shown any proof that they are real.*
2. *There must be intelligent extraterrestrial life in outer space, for no one has been able to prove that there isn't.*
3. *People have been trying for centuries to provide conclusive evidence for the claims of astrology, and no one has ever succeeded. Therefore, we must conclude that astrology is a lot of nonsense.*

Exceptions

1. if qualified researchers investigate a certain phenomenon within their range of expertise and fail to turn up any evidence that the phenomenon exists, this fruitless search by itself constitutes positive evidence about the question. It is not always necessary, however, that the investigators have *special* qualifications.
   The kinds of qualifications needed depend on the situation. Sometimes the mere ability to see and report what one sees is sufficient.
2. "Presumption of innocence" – "innocent until proven guilty"

11. HASTY GENERALIZATION

- It occurs when there is a reasonable likelihood that the sample is not representative of the group. Such a likelihood may arise if the sample is either too small or not randomly selected.

Examples:

- *Two weeks ago Gishen Pharmacy was robbed and the suspect was a street boy. Yesterday a street teenager snatched an old lady's purse while she was waiting at the corner bus stop. Clearly, homeless street boys are nothing but a pack of criminals.*
- *I had a terrible time with my former wife. I am never going to marry again because I have learned that no women are easy to live with.*

12. FALSE CAUSE

- It occurs whenever the link between premises and conclusion depends on some imagined causal connection that probably does not exist.
- There are three varieties:
- ***post hoc ergo propter hoc*** ("after this, therefore on account of this").
  - It presupposes that just because one event precedes another event, the first event causes the second.
- ***non causa pro causa*** ("not the cause for the cause").
  - It is committed when what is taken to be the cause of something is not really the cause at all and the mistake is based on something other than mere temporal succession.
- ***oversimplified cause***
  - It occurs when a multitude of causes is responsible for a certain effect but the arguer selects just one of these causes and represents it as if it were the sole cause.

13. SLIPPERY SLOPE

- It is a variety of false cause fallacy.
- It occurs when the conclusion of an argument rests on an alleged chain reaction and there is not sufficient reason to think that the chain reaction will actually take place.

Examples:

Attempts to outlaw pornography threaten basic civil rights and should be summarily abandoned. If pornography is outlawed, censorship of newspapers and news magazines is only a short step away. After that there will be censorship of textbooks, political speeches, and the content of lectures delivered by university professors. Complete mind control by the central government will be the inevitable result.

14. WEAK ANALOGY

- The fallacy of **weak analogy** is committed when the analogy is not strong enough to support the conclusion that is drawn.

Examples:

1. *Marriage is like a basketball game. Basket ballers regularly take time outs to reflect on the state of the game and thus couples should take time out from being with each other too.*
2. *If a person uses an axe continually it will be blunted. So it follows that, as the mind is a delicate instrument, if I use it too much, I will dull it.*

5.4. FALLACIES OF PRESUMPTION

- The premises presume what they purport to prove.
- What is presumption?

15. BEGGING THE QUESTION*(petitio principii)*

- presumes that the premises provide adequate support for the conclusion when in fact they do not.
- is committed whenever the arguer creates the illusion that inadequate premises provide adequate support for the conclusion.
- This is done by
  - leaving out a possibly false (shaky) key premise,
  - restating a possibly false premise as the conclusion, or
  - reasoning in a circle.
- arguments that beg the question are normally valid but unsound.

Examples

1. *Murder is morally wrong. This being the case, it follows that abortion is morally wrong.*
2. *If such actions were not illegal, then they would not be prohibited by the law.*
3. *Since the exercise of unrestricted freedom is dangerous it follows that it is harmful to allow people unfettered liberty.*
4. *I know that she loves me because she told me so and I can believe what she told me because she loves me.*
5. ***X**: "People cannot help what they do."*
   ***Y**: "Why not?"*
   ***X**: "Because they always follow the strongest motive."*
   ***Y**: "But what is the strongest motive?"*
   ***X**: "The one that people follow of course."*
6. **X**: God exists.
   Y: How do you know?
   X: It's written in the Holy Bible.
   Y: But what is the Bible?
   X: It is the word of God?

16. Complex Question

- is committed when two (or more) questions are asked in the guise of a single question and a single answer is then given to both of them.
- presumes that a question can be answered by a simple “yes,” “no,” or other brief answer when a more sophisticated answer is needed.

EXAMPLES:

- *Have you stopped cheating on exams?*
- *Where did you hide the marijuana you were smoking?*
- *Are you going to be a good little boy and eat your hamburger?*
- *Is Helen still telling lies?*
- *How long must I put up with your snotty behavior?*
- *When are you going to stop talking nonsense?*

17. False Dichotomy

- is committed when a disjunctive ("either . . . or . . .") premise presents two unlikely alternatives as if they were the only ones available, and the arguer then eliminates the undesirable alternative, leaving the desirable one as the conclusion.
- Such an argument is clearly valid, but since the disjunctive premise is false, or at least probably false, the argument is typically unsound.

Example

- *Either you let me attend Rophnan's concert or I'll hate you for the rest of my life. You don't want me to hate you for the rest of my life. Therefore, you'll let me attend the concert.*
- *Exceptions- when the alternatives are mutually exclusive*

18. Suppressed Evidence

- It does not fulfill the "total evidence" requirement for the premises ignore some important piece of evidence that outweighs the presented evidence and entails a very different conclusion.
- This fallacy is classified as a fallacy of presumption because it works by creating the presumption that the premises are both true and complete when in fact they are not.

FALLACIES OF AMBIGUITY

- These fallacies arise from the occurrence of some form of ambiguity in either the premises or the conclusion (or both).
- When the conclusion of an argument depends on a shift in meaning of an ambiguous word or phrase or on the wrong interpretation of an ambiguous statement, the argument commits a fallacy of ambiguity.

19. Equivocation

- occurs when the conclusion of an argument depends on the fact that a word or phrase is used, either explicitly or implicitly, in two different senses in the argument.
- Such arguments are either invalid or have a false premise, and in either case they are unsound.

Examples

1. *Some triangles are obtuse. Whatever is obtuse is ignorant. Therefore, some triangles are ignorant.*
2. *Any law can be repealed by the legislative authority. But the law of gravity is a law. Therefore, the law of gravity can be repealed by the legislative authority.*
3. *We have a duty to do what is right. We have a right to speak out in defense of the innocent. Therefore, we have a duty to speak out in defense of the innocent.*

20. Amphiboly

- occurs when the arguer misinterprets an ambiguous statement and then draws a conclusion based on this faulty interpretation. The original statement is usually asserted by someone other than the arguer, and the ambiguity usually arises from a mistake in grammar or punctuation—a missing comma, a dangling modifier, an ambiguous antecedent of a pronoun, or some other careless arrangement of words. Because of this ambiguity, the statement may be understood in two clearly distinguishable ways. The arguer typically selects the unintended interpretation and proceeds to draw a conclusion based on it.

Examples

1. *The tour guide said that standing in AAU, the obelisk could easily be seen. It follows that the obelisk is in AAU.*
2. *John told Henry that he had made a mistake. It follows that John has at least the courage to admit his own mistakes.*
3. *Professor Johnson said that he will give a lecture about heart failure in the biology lecture hall. It must be the case that a number of heart failures have occurred there recently.*
4. *Mrs. Hart stated in her will, "I leave my 500-carat diamond necklace and my pet chinchilla to Alice and Theresa." Therefore, we conclude that Alice gets the necklace and Theresa gets the chinchilla.*
5. *The king and the queen are visiting dignitaries. Therefore, the king and the queen are receiving dignitaries.*
6. *He only jogs at sunset.*

V. Grammatical Analogy

- Arguments that commit these fallacies are grammatically analogous to other arguments that are good in every respect. Because of this similarity in linguistic structure, such fallacious arguments may appear good yet be bad.
- They erroneously try to transfer attribute from whole to parts and vice versa.

21. Composition

- is committed when the conclusion of an argument depends on the erroneous transference of an attribute from the parts of something onto the whole. In other words, the fallacy occurs when it is argued that because the parts have a certain attribute, it follows that the whole has that attribute, too, and the situation is such that the attribute in question cannot be legitimately transferred from parts to whole.

22. Division

- Is committed when the conclusion of an argument depends on the erroneous transference of an attribute from a whole (or a class) onto its parts (or members)

Examples

1. *Each atom in this piece of chalk is invisible (has mass). So, the chalk is invisible (has mass).*
2. *Each player on this basketball team is an excellent athlete. Therefore, the team as a whole is excellent.*
3. *All the members of the association are professionals. So, it must be a professional association.*
4. *My little brother can tear each of the pages in the telephone book in two. So, he can tear the telephone book in two.*
5. *Selam has a very expensive collection of paintings. So, each of her paintings must be expensive.*
6. *America is a very rich country; so, all Americans are rich.*